



Module 19 Day 1: Splunk Dashboards and Visualizations

Overview

In today's class, you'll create dashboards that display useful visualizations of Splunk searches.

Class Objectives

By the end of class, you will be able to:

- Create visualizations of single- and multiple-value searches.
- Use the `geostats` and `iplocation` commands to add location-based visualizations to searches.
- Combine multiple visualizations in a single dashboard.
- Modify dashboards with time range input and drilldown capabilities.

Lab Environment

This class uses the Splunk instance inside the Ubuntu VM.

- To launch, move into the `/splunk` directory and run the following script:
 - `sudo ./splunk.sh`
 - On the first time running, select Option 1.
 - For all subsequent times, select Option 2.
- Once the script completes, pull up the web browser in the Ubuntu VM and enter the following URL:
 - `localhost:8000/`
- On the Splunk login page, use the following credentials:
 - User: `admin`
 - Password: `cybersecurity`
- The logs for each lesson and activity are located in the `/splunk/logs` directory.

Slideshow

The class slides are available on Google Drive here: [19.1 Slides](#)

01. Welcome Back and Module Recap

Today, you'll learn how security professionals use Splunk's visualization capabilities to identify and mitigate security issues.

Let's review what was covered in the last class:

- Splunk uses the Statistics feature to display specific values from search results in an easy-to-read spreadsheet.
- The `eval` command is used to design new fields or modify existing fields.
- Splunk uses reports to save searches and run them at designated times.
- Splunk alerts are designed to send automatic notifications when a specific condition, known as a trigger condition, is met.
- Designing poor alerts can cause false positives and false negatives.
 - False positives occur when conditions are met and an alert is triggered, but the security situation being monitored did not actually occur.
 - False negatives occur when the condition is met and an alert is not triggered, meaning the security situation occurred undetected.
- Baselining is a method of looking at historical data to determine how much activity is considered "normal."
- Security professionals use baselining to determine a threshold.
 - A threshold is a condition or level that, when met, triggers an alert.
- Security professionals may encounter the challenge of alert fatigue. This occurs when they receive so many alerts that they cannot adequately respond to each one.

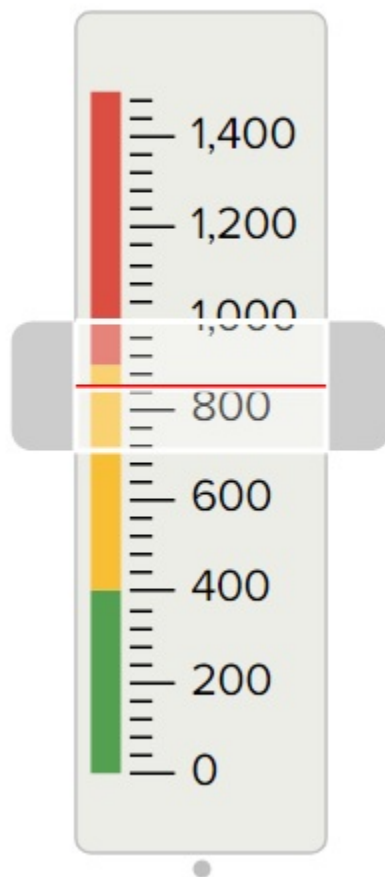
02. Introduction to Visualizations

Previous classes showed how to visualize statistics with charts. Today, you'll expand the Splunk capabilities to include adding more contextualized and informative visuals. These contribute to the analysis and research of system and security issues.

- For example, the table shows the number of logins into a web application per minute:

Events Patterns Statistics (1) Visualization	
20 Per Page ▼ Format Preview ▼	
TaskCategory ▾	total ▾
Logon	879

The gauge visualization contextualizes that number by including the severity of the login count:



Splunk uses visualizations to make complex data easier to understand and analyze:

- Splunk visualizations allow interactivity and offer more in-depth details.
- These visualizations range from simple bar and column charts to more complex horizon charts and punch cards.

Splunk visualizations can display single values, such as the total number of attacks, and multiple values, such as a chart of attacks correlated by attack type.

The following demonstration uses a radial gauge visualization.

- This demonstration uses event data from a Fortinet firewall attack log to create a radial gauge measuring the Oracle.9i.TNS.OneByte.DoS attack over a one-hour period.

- Specify a one-minute period by double-clicking the bar representing data from 4:24 p.m. (or another time of your choice).

- ## New Search

`source="fortinet_IPS_log.csv" host="cae59c5c10a8" sourcetype="csv"`

✓ 200 events (before 5/28/20 11:07:00 PM) No Event Sampling ▼

Events (200) Patterns Statistics Visualization

Format Timeline ▾ — Zoom Out + Zoom to Selection × Deselect

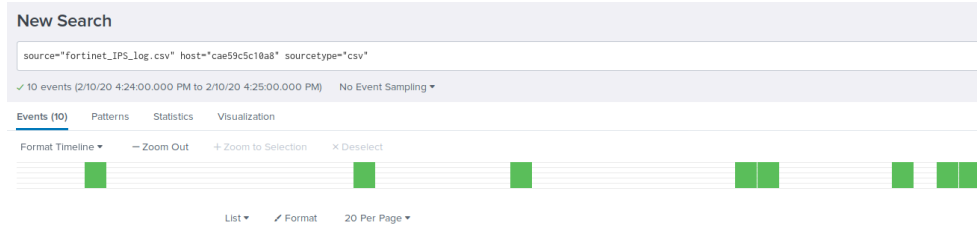
List ▾ ↗ Format 20 Per Page ▾

< Hide Fields	≡ All Fields	i Time	Event
SELECTED FIELDS		> 2/10/20 5:05:52.000 PM	"Feb 10 17:05:52 fortinet-01 date=2012-06-06,time=01:20:13,devname=FORTINET-01,device_id=FG300B390966 profilegroup="N/A",profiletype="N/A",profile="N/A",src=131.178.233.243,dst=91.214.92.22,src_int ed,proto=6,service=1521/tcp,vd="root",count=1,attack_name=Oracle.9i.TNS.OneByte.DoS,src_port=44419, 10725",user="N/A",group="N/A",incident_serialno=651335766,msg="DoS: Oracle.9i.TNS.OneByte.DoS" ",2020-02-10T17:05:52.000+0000,detected,,10725,Oracle.9i.TNS.OneByte.DoS,,N/A,DoS,,,DoS: Oracle.9i.TN 2.22,false,,,,untrust,1521,,false,,false,false,FG300B3909600791,FORTINET-01,,91.214.92.22,PCI-APP-C rtinet_fortinet_ips,detected,detected,signture,ips,N/A,127.0.0.1.0,network,NON-PCI-WEB,651335766,mai 6,,_:_:-=-=:,:,-=,=,=,=,=,"/" ,"/" ,"/" ,"/" ,"http://www.fortinet.com/ids/VID1072 rtinet,fortinet,prd-p-lttsvb2rkb18,,131.178.233.243,,,,,,,,NON-PCI-WEB,131.178.233.243,false,,,,unt

INTERESTING FIELDS

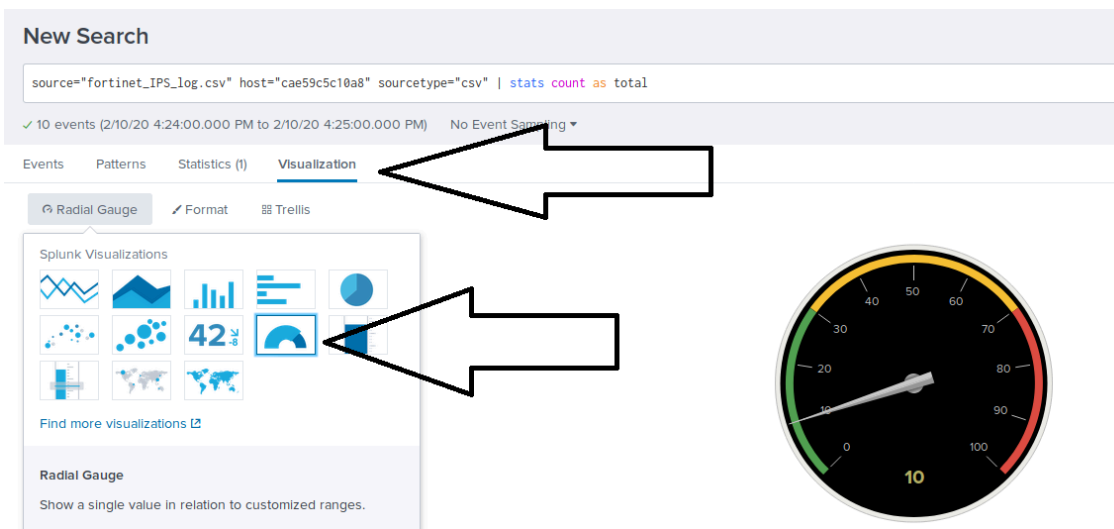
 - a action 1
 - # attack_id 1
 - a attack_name 1

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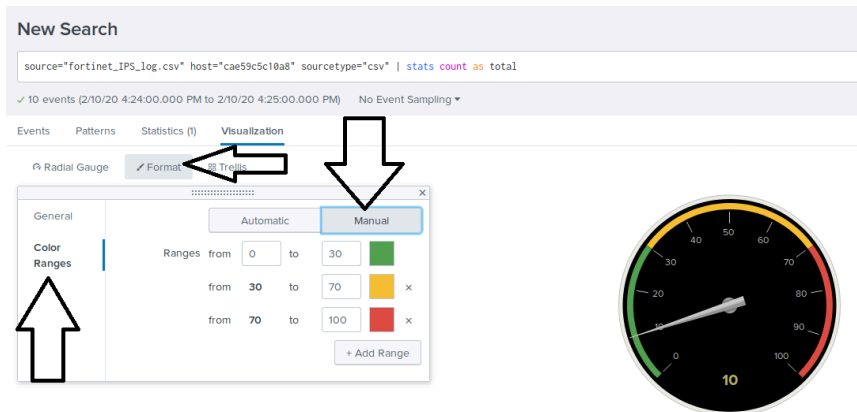


Note: Your results may vary.

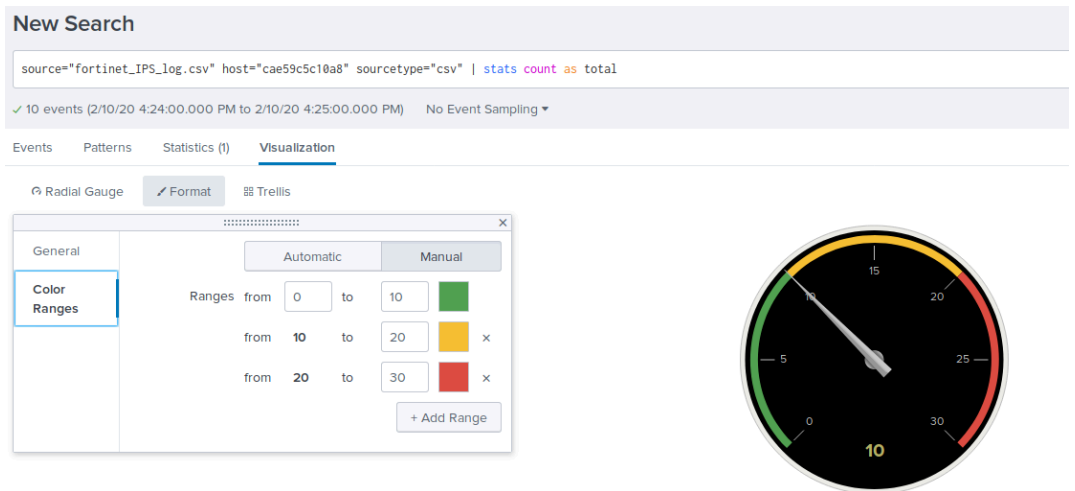
5. Next, turn this single value into a radial gauge.
 - Select **Visualization** and then the **Radial Gauge** option.



- Splunk created the different color values in this radial gauge by default:
 - Green: 0–30
 - Yellow: 30–70
 - Red: 70–100
6. Customize these ranges and colors based on your needs and preferences.
 - To edit, select Format > Color Ranges > Manual.



- In this example, the value of that minute is 10 events, which can be considered the top of a normal level.
- Change the ranges to:
 - Green: 0–10
 - Yellow: 10–20
 - Red: 20–30



- Now the radial gauge displays the 10 events at the top of the green range.
 - Save this radial gauge by selecting Save As > Report > Choose a Title.
 - Select the radial gauge icon under **Content**.

Save As Report



Your report has an absolute time range.

Title	Radial Gauge
Description	optional
Content	+
Time Range Picker	Yes No



Cancel

Save

03. Activity: Single-Value Visualizations

- [Activity File: Single-Value Visualizations](#)

04. Activity Review: Single-Value Visualizations

05. Multiple-Value Visualizations

We just created a visualization with a single value. Now, we'll use visualizations to display multiple values.

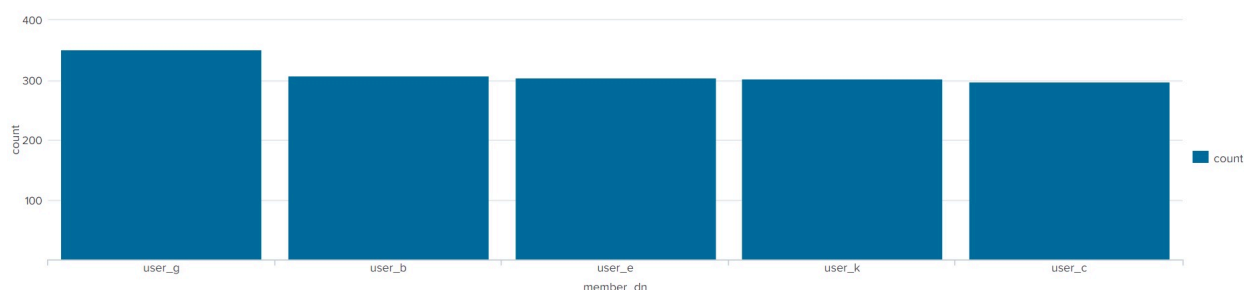
One scenario in which multiple value visualizations are useful is when a business experiences brute-force attacks against a web application and wants to visualize the list of users being attacked and the number of attacks experienced by each user.

These statistics will still be displayed in a spreadsheet:

member_dn	count
user_g	367
user_j	322
user_f	308
user_i	304
user_b	302
user_l	300
user_a	300

However, these spreadsheets can be converted into simple and interactive visualizations:

- For example, the data found in the spreadsheet can be displayed in a column chart.



- Note the interactivity: When the mouse is hovered over a bar, the number of attacks experienced by this user is displayed.



Multiple-Value Visualizations Walkthrough

In this demonstration, we'll use event data from Windows event logs to create a pie chart of top user logins.

- Load the following Splunk file: demo_winlogs.csv
- 1. Run the following search:

- source="demo_winlogs.csv" subject="An account was successfully logged on"
- Note: Your values and counts may differ from the screenshots.

New Search

source="wineventlog:security" subject="An account was successfully logged on"

✓ 758 events (3/6/20 3:54:45.000 PM to 3/6/20 4:09:45.000 PM) No Event Sampling ▼

Events (758) Patterns Statistics Visualization

Format Timeline ▼ — Zoom Out + Zoom to Selection × Deselect

2. Create a report of the top values.

- Click the User field and select Top values.

a src_user 53

a src_user_watchlist 1

a status 1

a subject 1

a ta_windows_action 1

a tag 8

a tag::action 1

a tag::app 2

a tag::eventtype 4

a TaskCategory 2

timeendpos 1

timestartpos 1

a Transited_Services 4

a Type 1

a user 17

a user_watchlist 1

a vendor 1

a Virtual Account 2

user

17 Values, 100% of events

Selected

Reports

Top values **Top values by time** Rare values

Events with this field

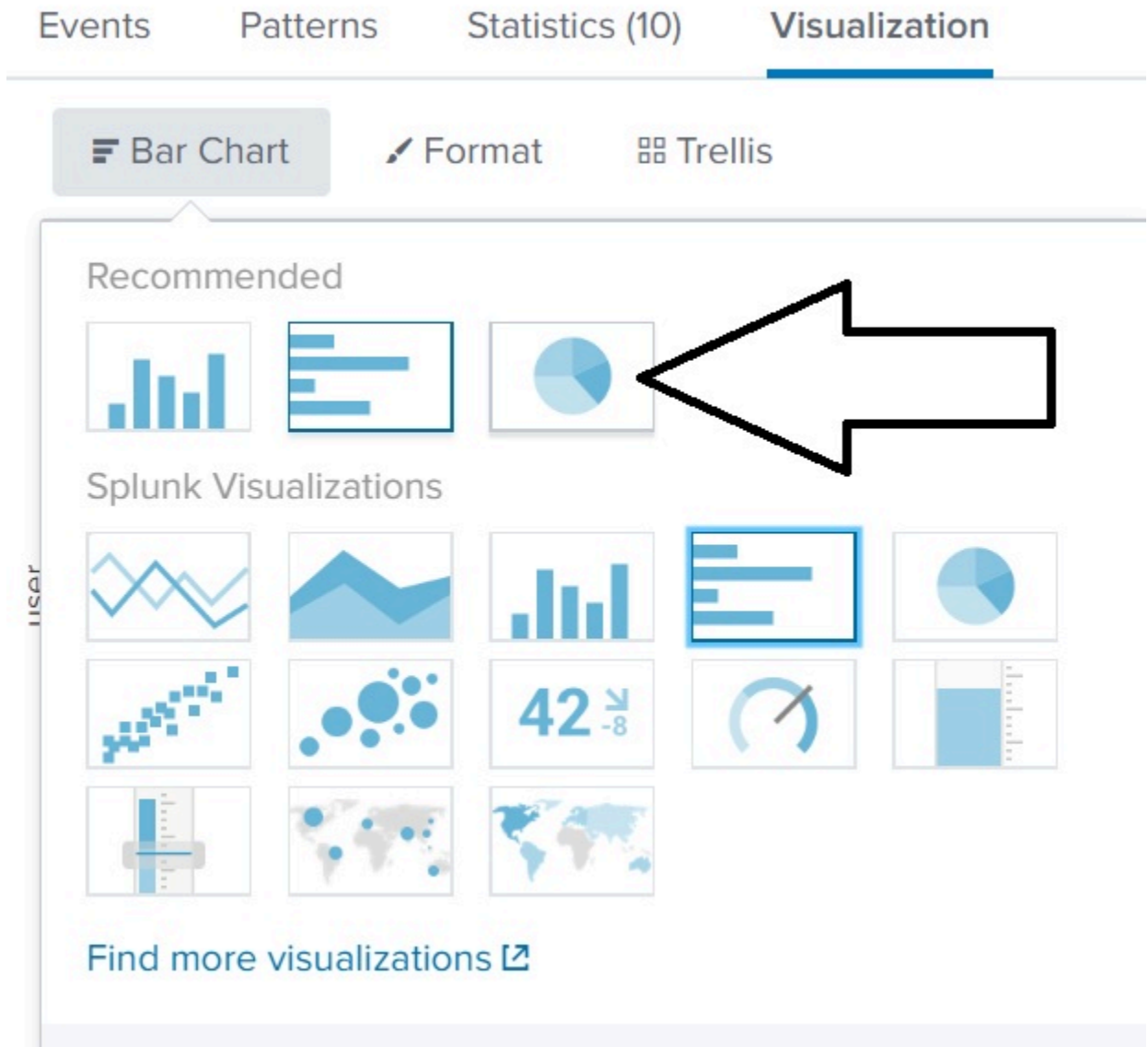
Top 10 Values	Count	%
user_g	64	8.377%
user_h	59	7.722%
user_a	55	7.199%
user_d	55	7.199%
user_l	55	7.199%
user_b	54	7.068%

3. Modify the top value to display the top 10 values.

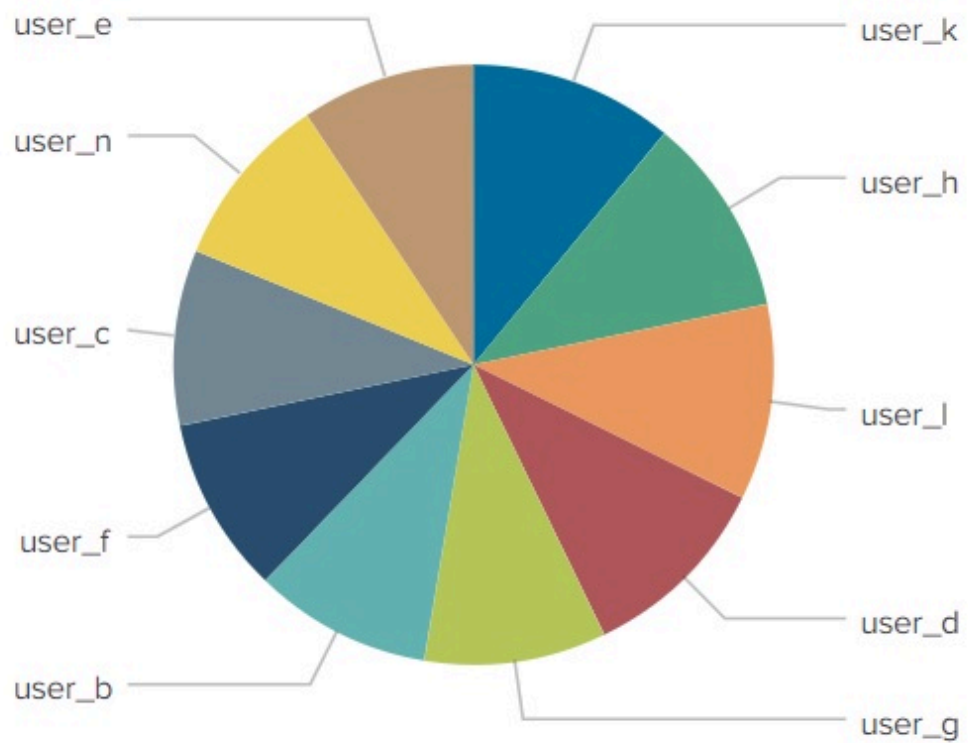
- Click Search to update the results.

source="wineventlog:security" subject="An account was successfully logged on" | top limit=10 user

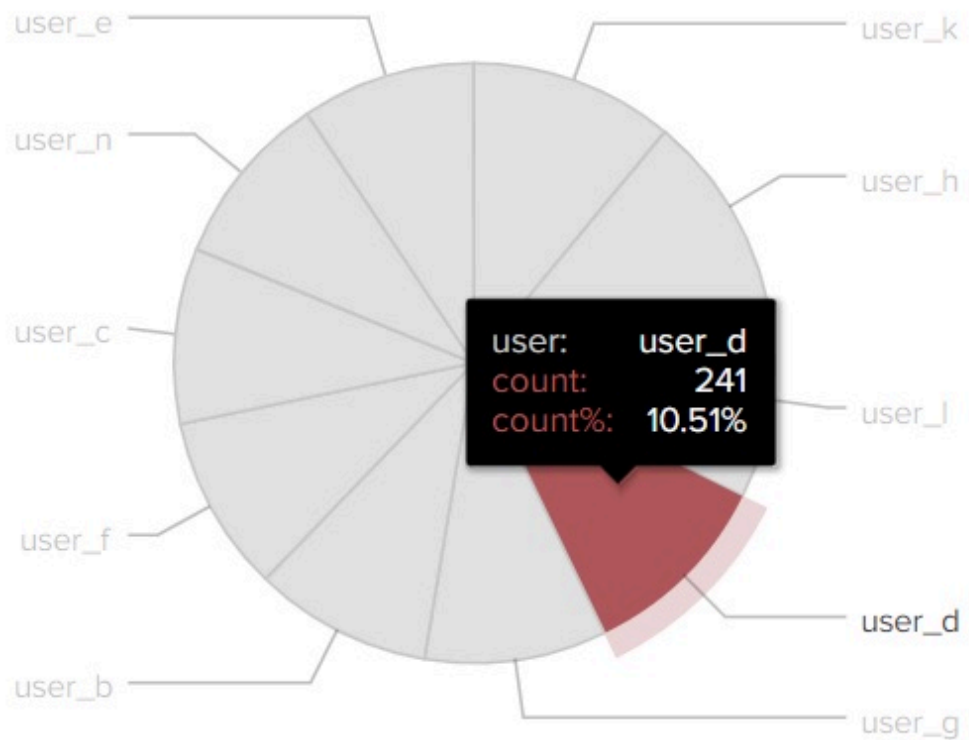
4. Under Visualization, select the pie chart icon under the Bar Chart heading.



- The top 10 users with login data are displayed as a pie chart.



5. Note the interactivity when moving your mouse over any piece of the pie.
 - It shows the specific count and percentage of a user.



6. Save the pie chart by selecting Save As > Report > Choose a Title.
 - Select the pie chart icon under Content.

06. Activity: Multiple-value Visualization

- [Activity File: Multiple-value Visualization](#)

07. Activity Review: Multiple-value Visualization

08. Geographic Map Visualizations

We can use visualizations to gain further insight into data, such as where an attack or activity originated.

- Organizations can monitor where users access the application from to determine the source of security issues.
- For example, a business's app users are primarily located in the United States. If the business is notified that a significant number of users have started accessing the application from another country, it is a cue to investigate the activity.

Splunk can create location-specific reports with the `iplocation` and `geostats` commands.

Geographic Map Visualization Walkthrough

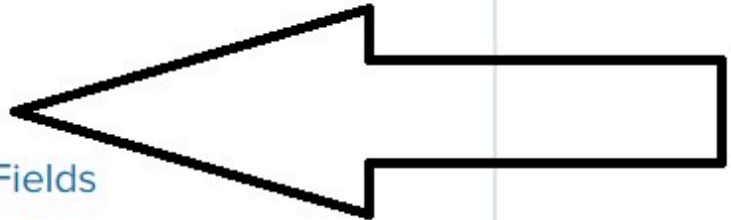
Load the following file: `demo_httplogs.csv`

1. Run the following SPL command in the Search & Reporting app:
 - `source="demo_httplogs.csv"`
 - This log shows HTTP activity.
2. Identify the field containing IP addresses. This is used to determine the location.
 - Use the source IP address, indicated by the field `src_ip`.
 - You can also use the destination IP (`dest_ip`).
 - To determine the location data of the IP, run the following SPL command:
 - `sourcetype="stream:http" | iplocation src_ip`
 - The `iplocation` command outputs the city and country data of the `src_ip` field.

- **Note:** To get the city and country results, you may have to select these fields from the more fields option below the field list.

a tag::eventtype 5
time_taken 5
timeendpos 1
a timestamp 100+
timestartpos 1
a transport 1
a uri_path 21
a uri_query 2
a url 100+

4 more fields
+ Extract New Fields



3. Search for the City and Country fields.

- Check the box to add the fields.

Select Fields

Select All Within Filter Deselect All Coverage: 1% or more ▼

City

i	✓	Field
>	☒	City



- The city and country should now appear on the Selected Fields list.

The screenshot shows a data visualization interface. On the left, the 'Selected Fields' list contains: `a City 100`, `a Country 50` (highlighted with a red arrow), `a host 2`, `a source 1`, and `a sourcetype 1`. Below this is the 'Interesting Fields' list with various fields like `action`, `app`, `bytes`, `bytes_in`, `bytes_out`, `cookie`, `date_hour`, `date_mday`, and `date_minute`. On the right, the 'Country' field details panel is open, showing '50 Values, 70.544% of events' and a 'Selected' toggle set to 'Yes'. Below this are tabs for 'Reports' (Top values, Top values by time, Rare values) and 'Events with this field'. The 'Top 10 Values' table is displayed:

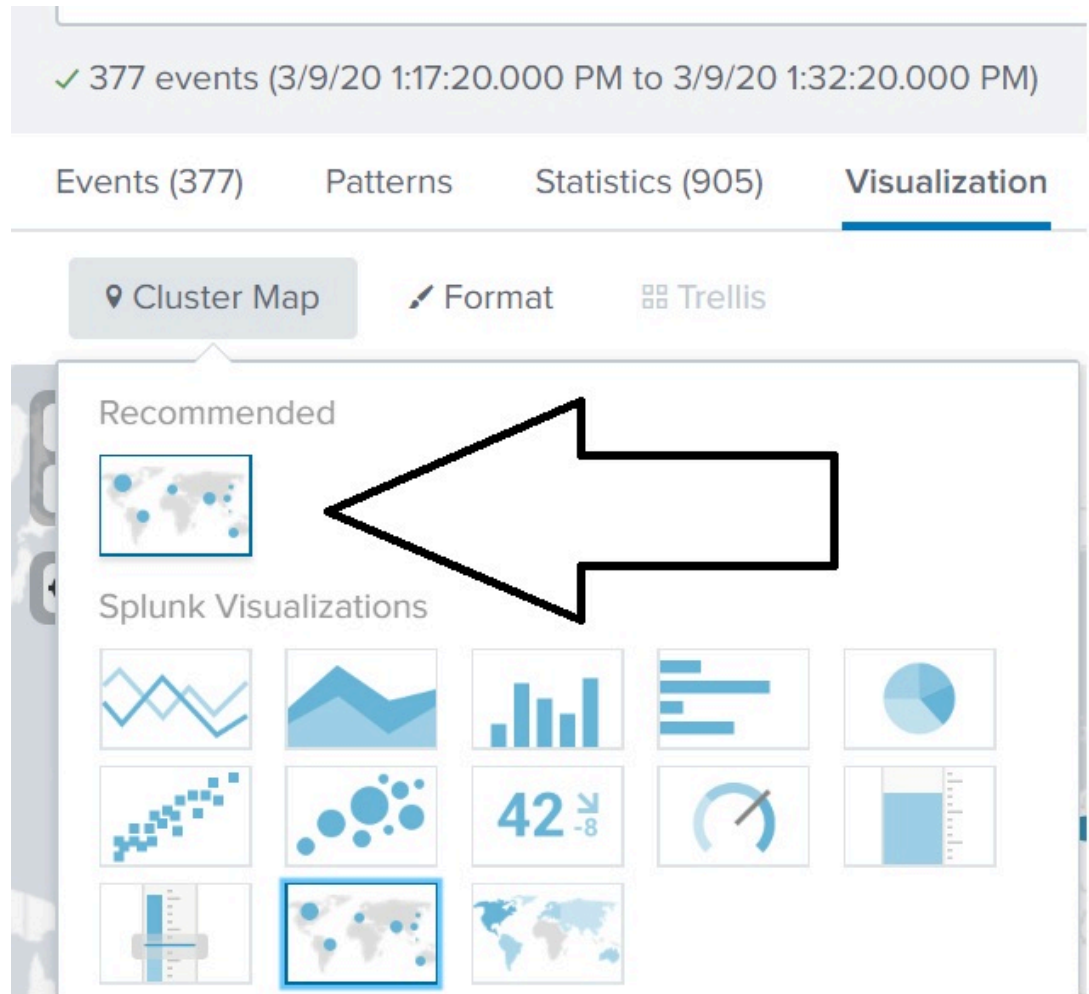
Top 10 Values	Count	%
United States	108	37.895%
China	38	13.333%
Japan	13	4.561%
France	11	3.86%
Germany	9	3.158%
South Korea	9	3.158%
United Kingdom	9	3.158%
Brazil	7	2.456%
Italy	6	2.105%
Spain	6	2.105%

- Click on the Country field to list the countries found based on the source IP address.

The `geostats` command uses the location data found with the `iplocation` command to map latitude and longitude data for each event.

1. Use the `geostats` command by running the following search:
 - `source="demo_httplogs.csv" | iplocation src_ip | geostats count`
2. Next, go to the Visualization tab.

- Splunk has recommended a visualization called Cluster Map based on your search.

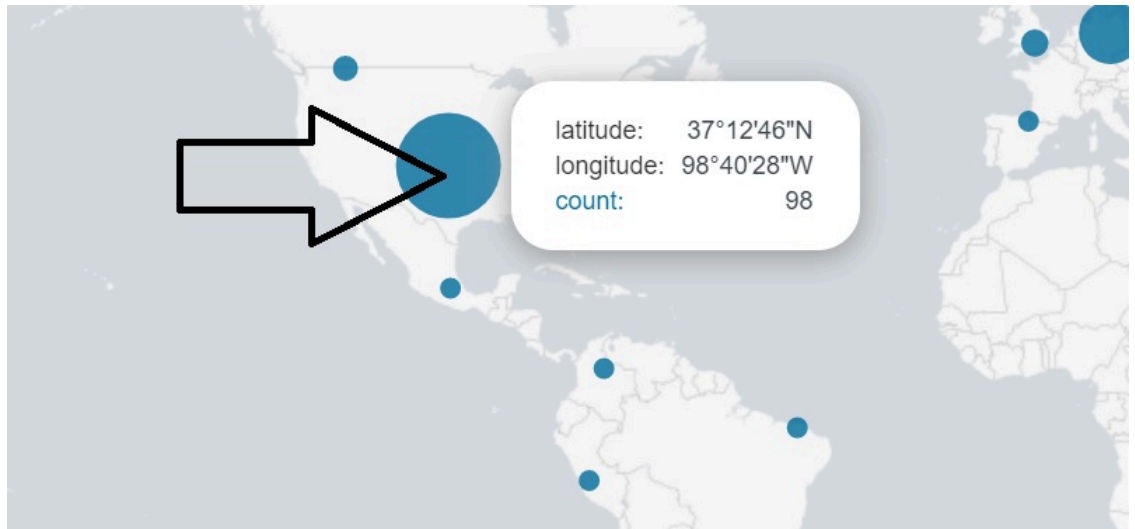


- Select the cluster map icon. This tool places the search results on a geographic map.

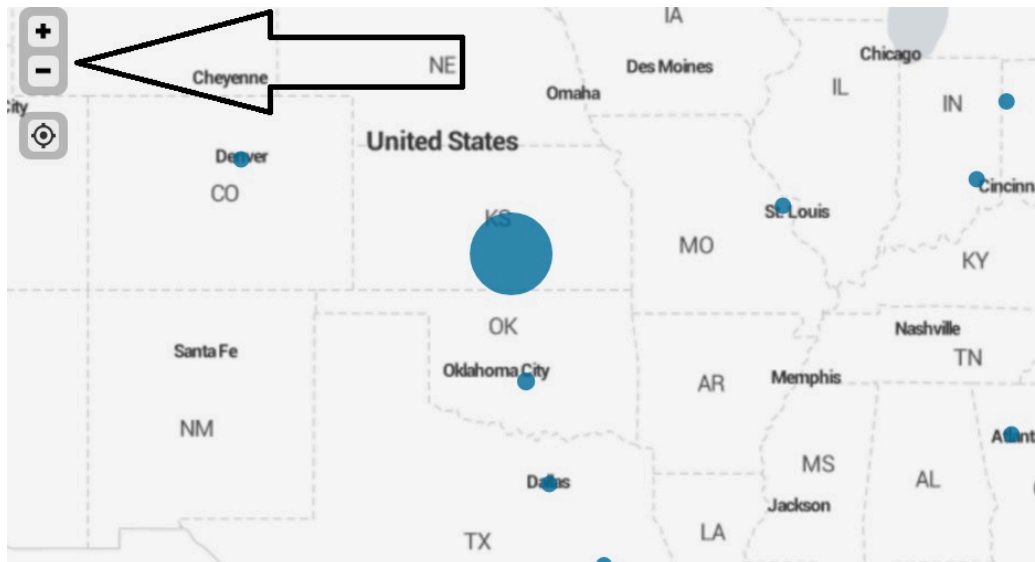


3. Hover over each circle to see the interactivity that displays the latitude, longitude, and count.

- The size of each circle indicates the number of events in that city.



- Use the + and - buttons on the left of the map to zoom in and out.



In summary, geographic maps in Splunk are powerful visualizations that help security teams quickly understand the origin of activity.

09. Activity: Geographic Map Visualization

- [Activity File: Geographic Map Visualization](#)
-

10. Activity Review: Geographic Map Visualization

11. Break

12. Introduction to Dashboards

So far, we've covered the following visualizations:

- Single value
- Multiple values
- Geographic maps

While each visualization is useful on its own, they are even more effective when grouped and displayed together.

For example, an organization may want to view all of the following simultaneously:

- The volume of successful and unsuccessful logins on the website
- A geographic map of the sources of activity
- A pie chart of the specific website pages accessed

This provides the security analyst with a complete picture of the state of their web application, using a Splunk tool called dashboards.

- Dashboards are a collection of multiple visualizations in a single location.
- The visualizations are placed in different sections, called panels.
- Panels can contain:
 - Single-value visualizations
 - Multiple-value visualizations
 - Geographic maps
 - Statistical charts

- The SOC team often has dashboards displayed on multiple screens in its operations room to provide availability and functionality across the staff.



Creating Dashboards Demonstration

Create a dashboard using the following scenario:

- As SOC manager, you want to create a single three-panel dashboard to monitor your Windows server. You want the panels to include:
 - A radial gauge of successful logins
 - A pie chart of users logging in
 - A statistical chart of the data in the pie chart

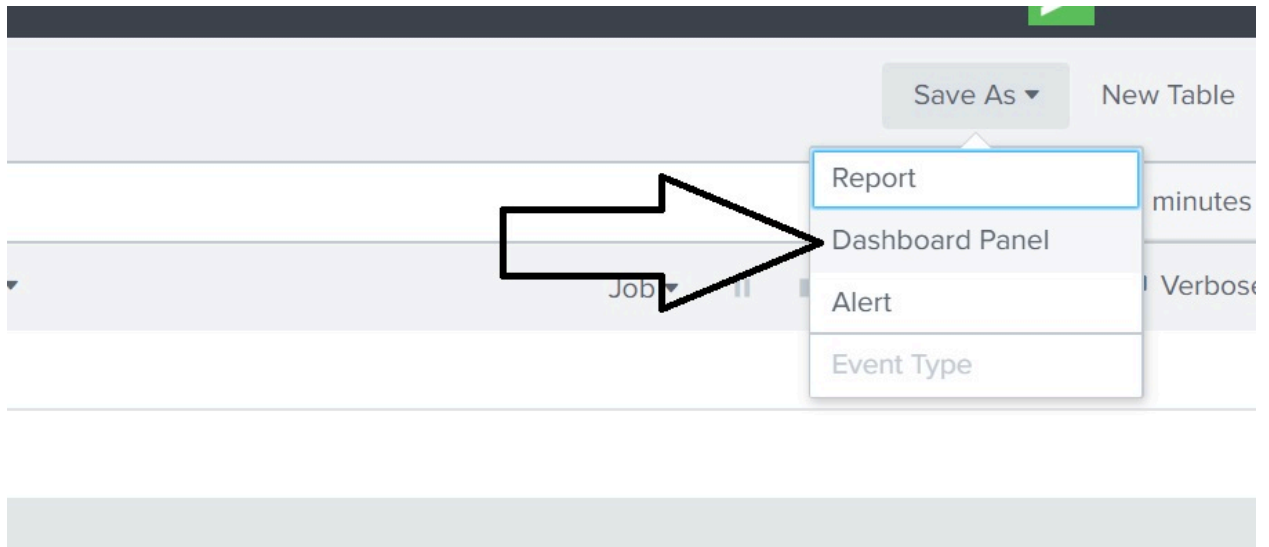
Note: Use a log file for this demonstration that has already been used: `demo_winlogs.csv`

- There is no need to reload this file.

Create each visualization, starting with the radial gauge.

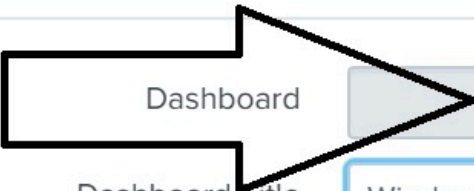
1. Add the radial gauge with the following search:
 - `source="demo_winlogs.csv" signature="An account was successfully logged on" | stats count as total`
 - Design and format the radial gauge visualization as needed.

2. To add this radial gauge to a dashboard, select Save As > Dashboard Panel.



3. Since you haven't created the dashboard yet, select the New Dashboard option.
 - o Name the dashboard "Windows Monitoring Dashboard."

Save As Dashboard Panel



Dashboard ☒ New ☐ Existing

Dashboard Title

Dashboard ID [?]
Can only contain letters, numbers and underscores.

Dashboard Description

Dashboard Permissions ☒ Private ☐ Shared in App

Panel Title

- Name the panel "Windows Login Count."
- Select Radial Gauge under Panel Content.

Save As Dashboard Panel

Dashboard Description

optional

Dashboard Permissions

Private

Shared in App

Panel Title

Windows Login Count

Panel Powered By ?

Q Inline Search

Drilldown ?

No action

Panel Content

Statistics

Radial Gauge

Cancel

Save

- Select Save.
4. Confirm that the dashboard has been created by selecting View Dashboard.
- This dashboard has only one panel. Let's add the next panel.

Your Dashboard Panel Has Been Created



The panel has been created and added to windows_monitoring_dashboard. You may now view the dashboard.



Windows Monitoring Dashboard

Windows Login Count



5. Next, add the pie chart.
 - Return to the Search & Reporting app and add the following search:
 - `source="demo_winlogs.csv"| top limit=10 user`
6. Select the pie chart as the visualization. Then select Save As > Dashboard Panel again.

Since you already created the dashboard, you just need to add the new visualization to it.

 - Select Existing > Windows Monitoring Dashboard.

Save As Dashboard Panel



Dashboard ☐ New ☒ Existing

Integrity Check of Installed Files ▼

Panel Title

Panel Powered By ☒ Integrity Check of Installed Files

Drilldown ?

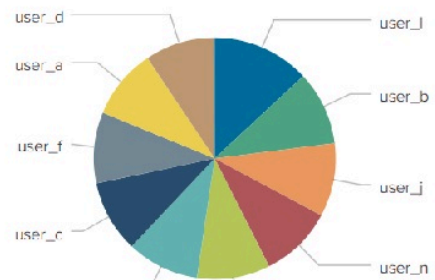
Panel Content ☐ Statistics ☒ Pie Chart

- Title the dashboard panel "Top 10 Windows Users."
- Select Save > View Dashboard.
- Now the dashboard has multiple visualizations.

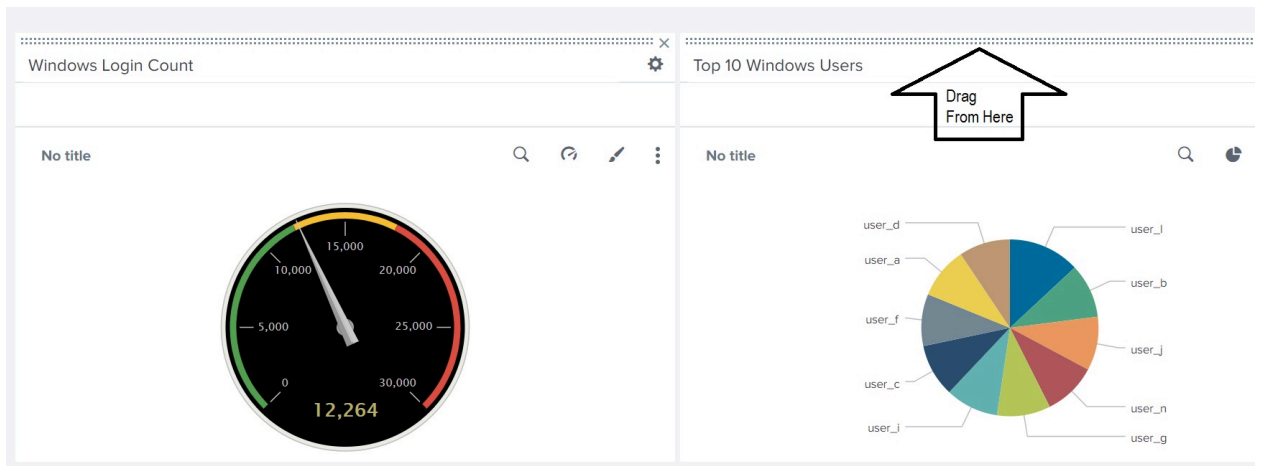
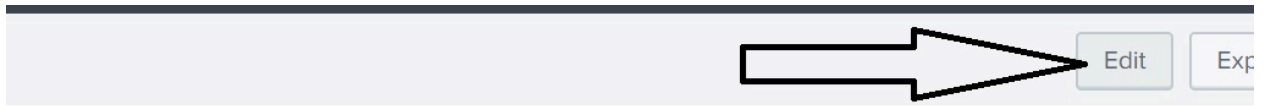
Windows Login Count



Top 10 Windows Users

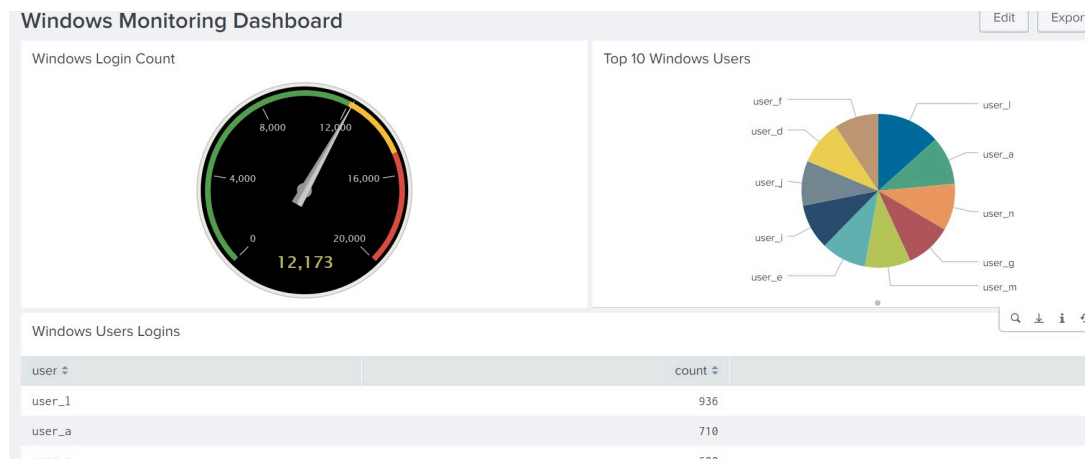


7. Dashboards can be edited by selecting Edit.
 - Rearrange panels by dragging the panel to a new location on the dashboard.



- Select Save to save your changes to the dashboard.
- 8. The last step is to add the third panel to the dashboard.
 - Add the search:
 - `source="demo_winlogs.csv" | top limit=10 user`
 - Follow the same steps for adding this to the dashboard.

- The completed dashboard should look like the following:



13. Activity: Creating Dashboards

- [Activity File: Creating Dashboards](#)

14. Activity Review: Creating Dashboards

15. Drilldowns and Dashboard Interactivity

Similar to many other applications in Splunk, dashboards have advanced features that help information security professionals research security issues.

These include:

- Direct input modifications within dashboards
- Drilldowns within panels for further analysis

Configure these features by using the dashboard and scenario from the preceding section:

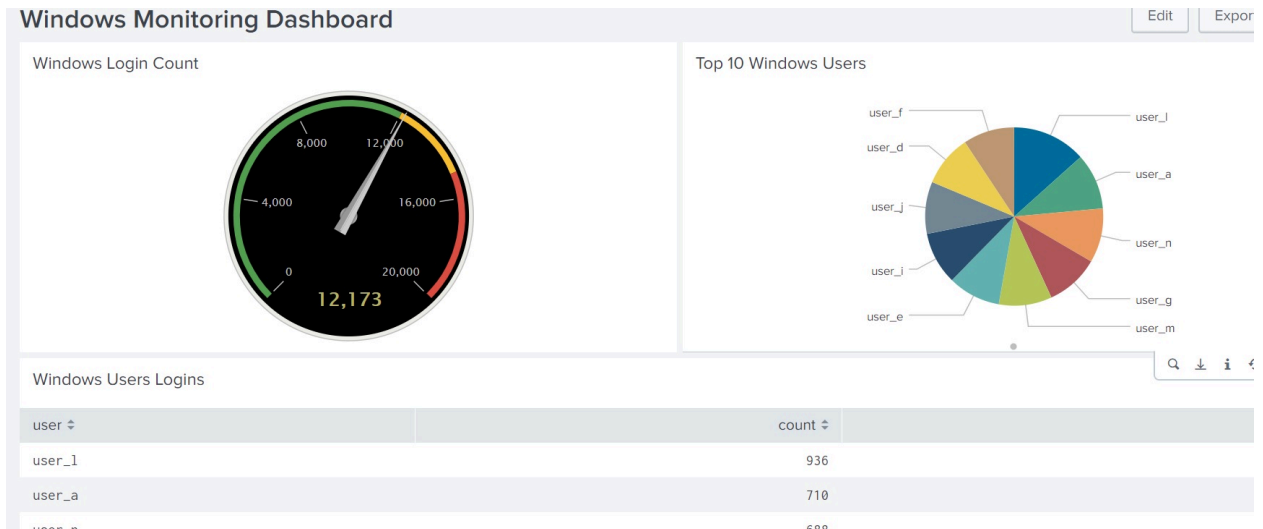
- As SOC manager, you created a three-panel dashboard to monitor your Windows server.
- Now expand the functionality of this dashboard by:
 - Modifying the date and time ranges being analyzed directly on the dashboard;
 - and

- Adding a drilldown into the visualizations to assist with further analysis.

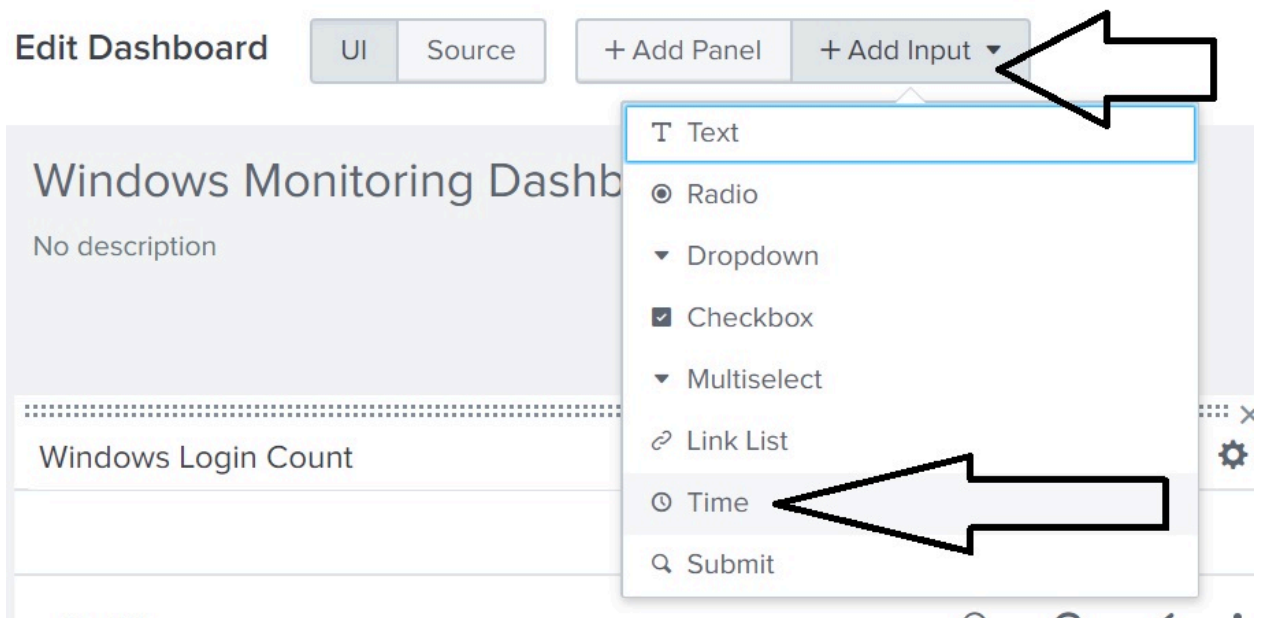
Time Input Walkthrough

First, configure the dashboard to modify the date and time range.

1. Open the dashboard created in the preceding demonstration:
 - Select Dashboards > Windows Monitoring Dashboard.

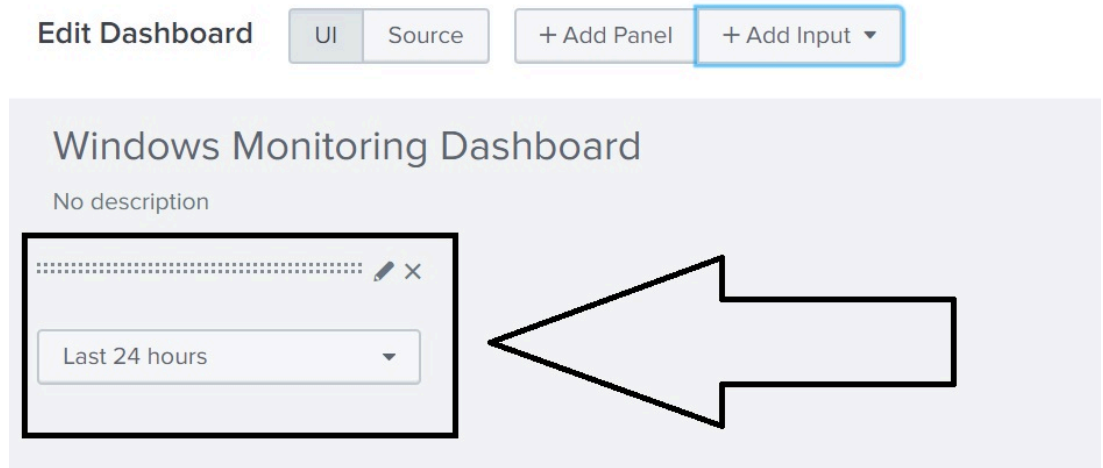


2. Select Edit to modify the dashboard.
 - Select Add Input, then select Time.
 - You can modify many features in the dashboard. Today, we'll only cover the Time feature.

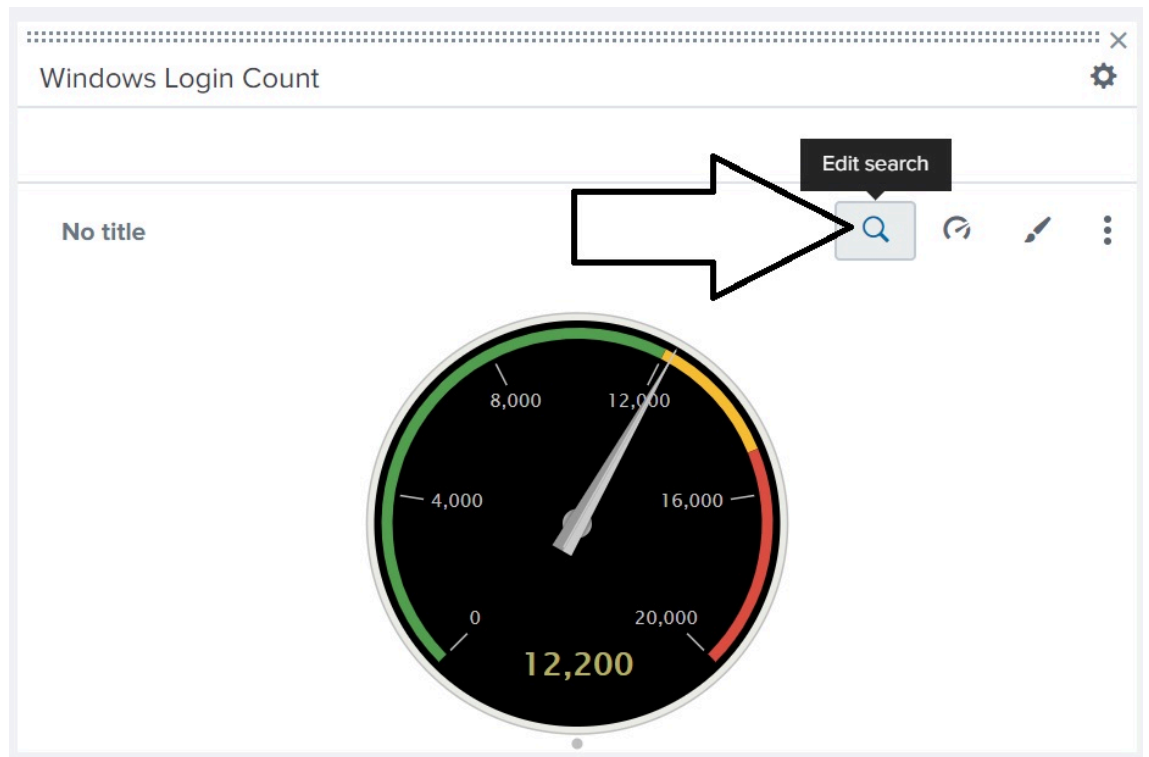


3. The top of the dashboard should now show a field where you can input time.

- This feature modifies the time range of data displayed in each panel.



4. For the panels to work with this new feature, reconfigure some settings for each panel.
 - To modify the first panel, select the magnifying glass icon to edit the search for the radial gauge.



- Select Shared Time Picker.
 - This is the input just created.
- Select Apply.

Edit Search ✕

Title

Search String `source="wineventlog:security" | stats count as total`

[Run Search](#)

Time Range Shared Time Picker (field1) ▼

Auto Refresh Delay ? ✓ Shared Time Picker (field1)

Refresh Indicator Use time picker

Tokens

Global

Cancel Convert to Report Apply

- Repeat these steps for the other two panels.
- Select Save to save the whole dashboard.
- There is now a working input on the dashboard where you can quickly change the time range of each panel.



Drilldown Walkthrough

Next, add a drilldown to help research specific data from a panel.

A drilldown is similar to a drop-down menu and has several functions.

1. Select the Edit option to modify the dashboard.
2. On the pie chart, select the three vertical dots for more actions, then select Edit Drilldowns.
 - Select Link to search in the drop-down.

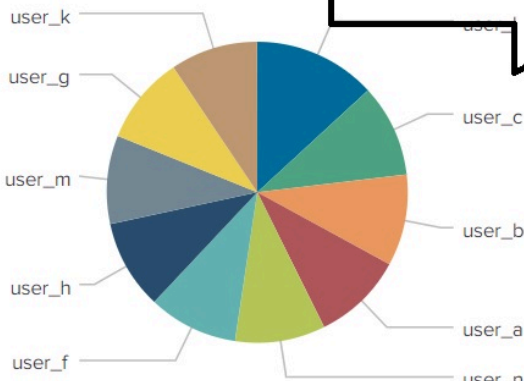
- Keep Auto selected.
- Click Apply.

Top 10 Windows Users

Window

No title

No title



user_k

user_g

user_m

user_h

user_f

user_n

user_a

user_b

user_c

user_d

Edit Drilldown

Trellis

Drilldown Editor

×

On Click

Link to search

Auto

Custom

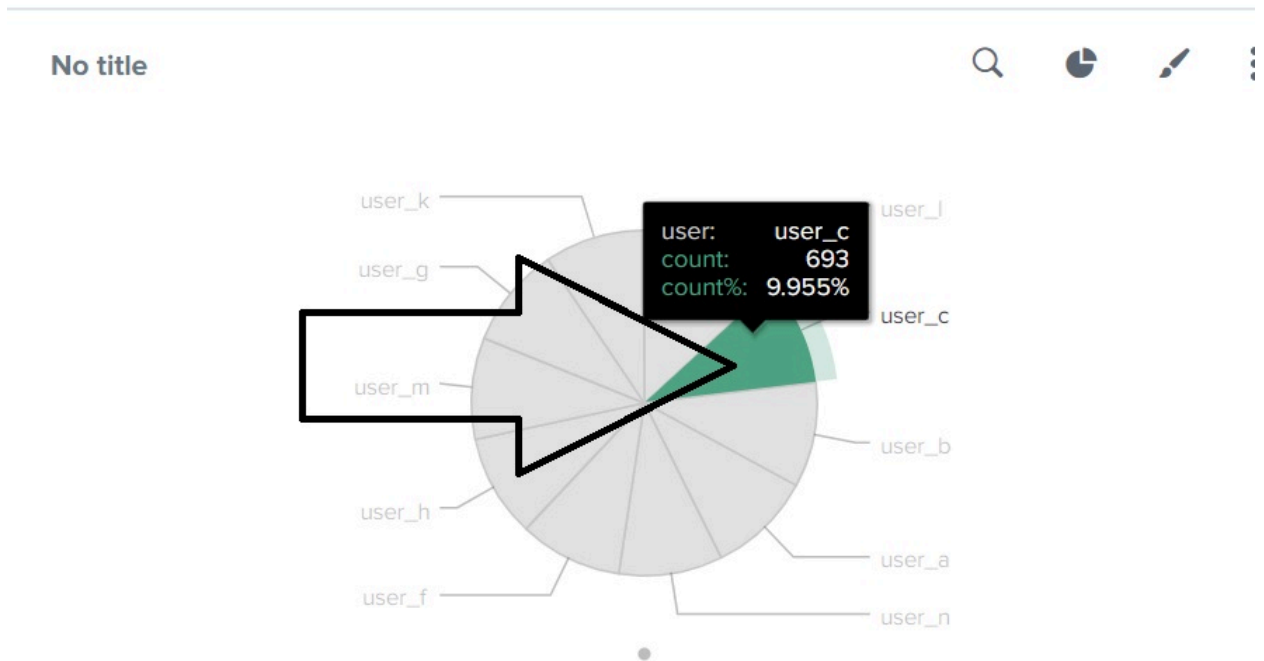
A search generates automatically using values from the clicked element.

Cancel

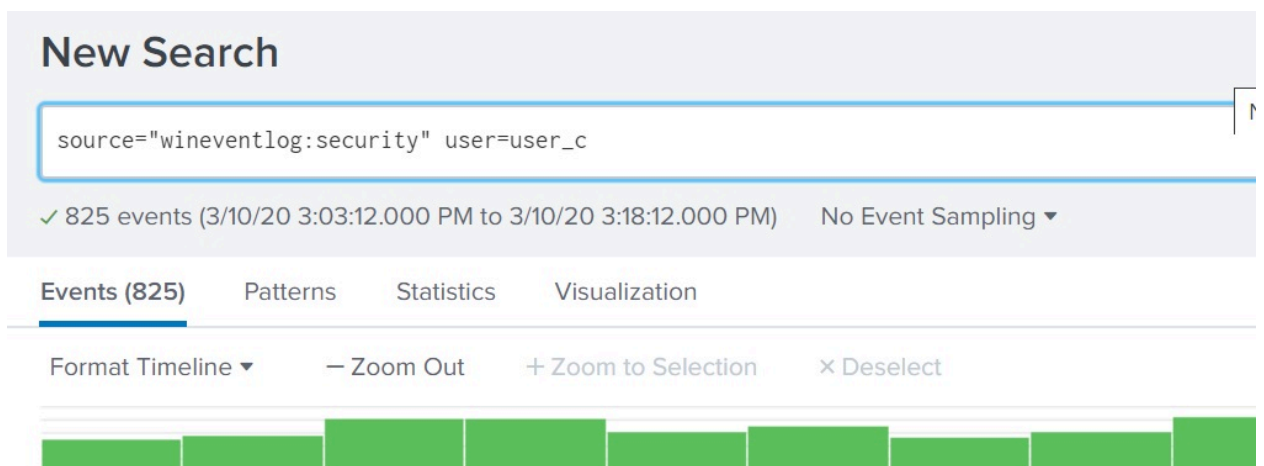
Apply

- Save the dashboard.

3. You can now search for a specific piece of the pie chart.
- Select one of the pie chart pieces.



- This creates a search based on what was selected.



This drilldown feature allows SOC analysts to analyze data directly from the dashboard.

16. Activity: Advanced Dashboards

- [Activity File: Advanced Dashboards](#)

17. Activity Review: Advanced Dashboards

18. Additional Knowledge: `table` command

The `table` command:

- The `table` command in Splunk is a non-streaming command that returns a table formed by only the fields that you specify in the arguments. The columns are displayed in the same order that fields are specified, and the column headers are the field names. Each row in the table represents an event, and the rows are the field values.
- The `table` command retrieves the raw data found in the base search, but only the data within the specified fields. It then places this data in a tabular format.
- The `table` command can also be used in conjunction with other commands for more complex searches. For example, you can use the `table` and `eval` commands together for faster searches. Similarly, the `stats` and `table` commands can be used together, but the `table` command results will be limited because the `stats` command is a transforming command.

It's important to note that the `table` command does not allow you to rename fields, only to define the fields you want to display in your tabulated results. If you need to rename a field, do it before the results are piped to the `table` command. Also, the `table` command truncates the number of results returned based on the settings in the `limits.conf` file.